

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (AIML) (Credit System) Degree Examinations**

January 2023

20AM516 – IMAGE AND VIDEO PROCESSING

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I

Unit – I

	Marks	BT*	CO*	PO*
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1.
 - a) With an example explain feature recognition and extraction.

	6	L*2	1	1
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 - b) What is Image augmentation? Why is it needed? Mention few Applications.

	6	L2	1	1
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 - c) Explain Translation, Scaling, Rotation, Shearing transformations with suitable example.

	8	L2	1	1
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2.
 - a) With an example explain Background subtraction in a video.

	6	L2	2	1
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 - b) What are the different levels of image processing? Mention different types of images.

	6	L1	2	1
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 - c) Perform the Histogram Equalization for image shown below.

Gray Level	0	1	2	3	4	5	6	7
No. of Pixel	11	9	15	5	6	8	6	4

	8	L2	2	1
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3.
 - a) Why do you need to perform enhancement? How is it done? Briefly explain its types.

	10	L2	1	1,2
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 - b) Briefly explain the fundamental steps in Digital Image Processing.

	10	L2	1	1,2
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Unit – II

4. a) Explain the following terms with example: i) Erosion ii) Closing in morphological image processing.	10	L2	4	1,2
b) How do you perform the following morphological operations? i) Boundary extraction ii) Region filling iii) Convex hull	10	L3	4	1,2
5. a) Explain following terms with example: i) Opening ii) Dilation in morphological image processing.	10	L3	4	1,2
b) How do you perform template matching with hit or miss transformation with suitable examples.	10	L3	4	1,2
6. a) Explain the RGB Model with the neat diagram.	10	L2	3	1,2
b) What is Pseudo color image processing? Explain intensity slicing Technique with neat Diagram.	10	L2	3	1,2

Unit – III

7. a) With an example explain i) Gaussian noise ii) Rayleigh noise iii) Gamma noise	10	L2	5	1,2
b) Explain restoration using Wiener filters.	10	L2	5	1,2
8. a) Explain restoration using Inverse filters.	10	L2	5	1,2
b) With an example explain Degradation/Restoration model.	10	L2	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
